MILITARY SPECIFICATION

For

AMPLIFIER, AUDIO FREQUENCY

AM-1780A/VRC

This specification is approved for use by Communication-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the AM-1780A/VRC Amplifier, Audio Frequency (see 6.1) for use in vehicular communications systems utilizing the radio set AN/VRC-12( ) units (see 6.1 and 6.8).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: Commander, U.S. Army Communication Electronics Command and Fort Monmouth, ATTN: DRSEL-ED-TO, Fort Monmouth, New Jersey 07703, by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5820

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2. APPLICABLE DOCUMENTS

2.1 Government Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall be considered a superseding requirement.

SPECIFICATIONS:

Military	
MIL-P-116	Preservation, Methods of
MIL-S-901	Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-I-6869	Impregnants for Aluminum Alloys and Magnesium Alloys Castings
MIL-P-11268	Parts, Materials and Processes used in Electronic Equipment
MIL-M-13231	Marking of Electrical Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-P-50884	Printed Wiring Flexible Printed Wiring
MIL-P-55110	Printed Wiring Boards
MIL-C-55543	Cable, Electrical Flat Multi-conductor, Flexible, Unshielded
MIL-E-55585	Electronic Equipment and Parts, Packaging of
Appendix B	* U.S. Army Nuclear and Chemical Agency - Nuclear Survivability Criteria for the Product Improved AM-1780A/VRC, 30 June 1982

STANDARDS:

Military	
MIL-STD-105	Sampling Procedures and Tables for Inspections by Attributes
MIL-STD-109	Quality Assurance Terms and Definition
MIL-STD-252	Wired Equipment, Classification Visual and Mechanical Defect
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-276	Impregnation of Porous Non-ferrous Metal Castings

* This document is classified and can be obtained from the procuring activity.

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MIL-STD-280	Definitions of Item Levels, Item Exchangeability, Models and Related Terms
MIL-STD-454	Standard General Requirement for Electronic Equipment
MIL-STD-461	Electromagnetic Characteristics for Electronic Equipment
MIL-STD-462	Electronic Interference Characteristics, Measure of
MIL-STD-726	Packaging Requirements Code
MIL-STD-781	Reliability Test Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1275	Electrical Circuits, 28 Volts DC Transient Characteristics for Military Vehicles

DRAWINGS

SC-GL-323191	Gages for Amplifier AM-1780/VRC (PCB gages excluded)
SC-A-46439	List of Accessories for Package Tester
SC-D-48664	Bracket Assembly
SM-D-415291	AM-1780A/VRC Case Drawing
SM-D-415294	AM-1780A/VRC Case Cover Drawing
SM-D-991090	Amplifier, Audio Frequency AM-1780A/VRC
SM-A-991154	Test Requirements Specification - Audio Amplifier PWB
SM-A-991155	Test Requirements Specification - DC-DC Converter PWB

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of Precedence. In the event of conflict, the contract shall have precedence over the requirements of this specification. This specification has precedence over the requirements of documents referenced herein.

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3. REQUIREMENTS

3.1 Description. This item is a direct replacement for the AM-1780/VRC Amplifier, Audio Frequency unit. The AM-1780A/VRC provides a switching function to allow crew members of a vehicle to communicate with each other and over vehicle radio sets. Functionally the AM-1780A/VRC controls keying of the radio transceivers by the crew, implements the switching necessary to route audio signals from other crew members or radios to the control box for earphone listening, and contains circuits for obtaining power from a radio or other power source (see Figure 1 and 2 for interconnections). Modular construction and printed wiring are employed in the Audio Amplifier. The amplifier operates from a 26 volt direct current (dc) nominal input.

3.1.1 Main Power Switch. This switch controls application of DC power to the AM-1780A/VRC circuit breaker.

3.1.2 Power Circuit Breaker. This switch controls application of DC power to AM-1780A/VRC circuits and peripheral crew boxes. It also provides overload protection of 1.5 amps.

3.1.3 Power Indicator. A green indicator lamp shall be provided to indicate when power circuit breaker switch is on. When lens cap is rotated one-quarter counterclockwise, the light does not shine through the lens.

3.1.4 Intercom Accentuated Switch. When in the on position, this switch shall reduce the audio level from the radio receivers whenever the intercom control is energized.

3.1.5 Radio Transmission Selector Switch. This switch controls the transmitter keying lines of the radios as follows:

a) Commander and Crew (Position 1). All crewmembers can key receiver-transmitters.

b) Commander Only (Position 2). Only the crew control unit connected to J504 may initiate radio transmissions.

c) Listening Silence (Position 3). No crew control unit shall be permitted to initiate radio transmissions.

3.1.6 Installation Switch. This switch serves several purposes.

. Controls terminations of audio signals.

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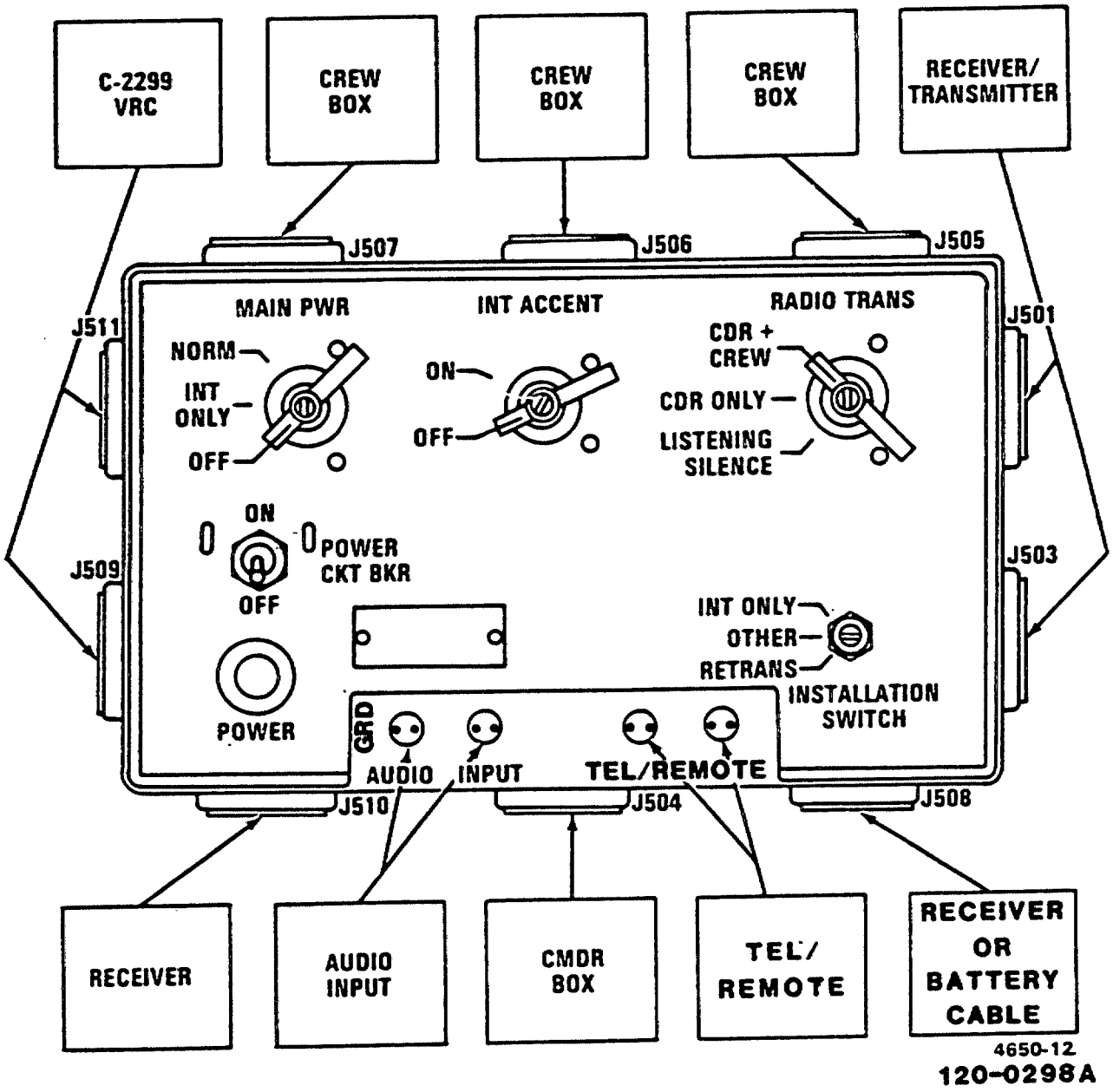


Figure 1 AM-1780A/VRC SYSTEM INTERCONNECTIONS

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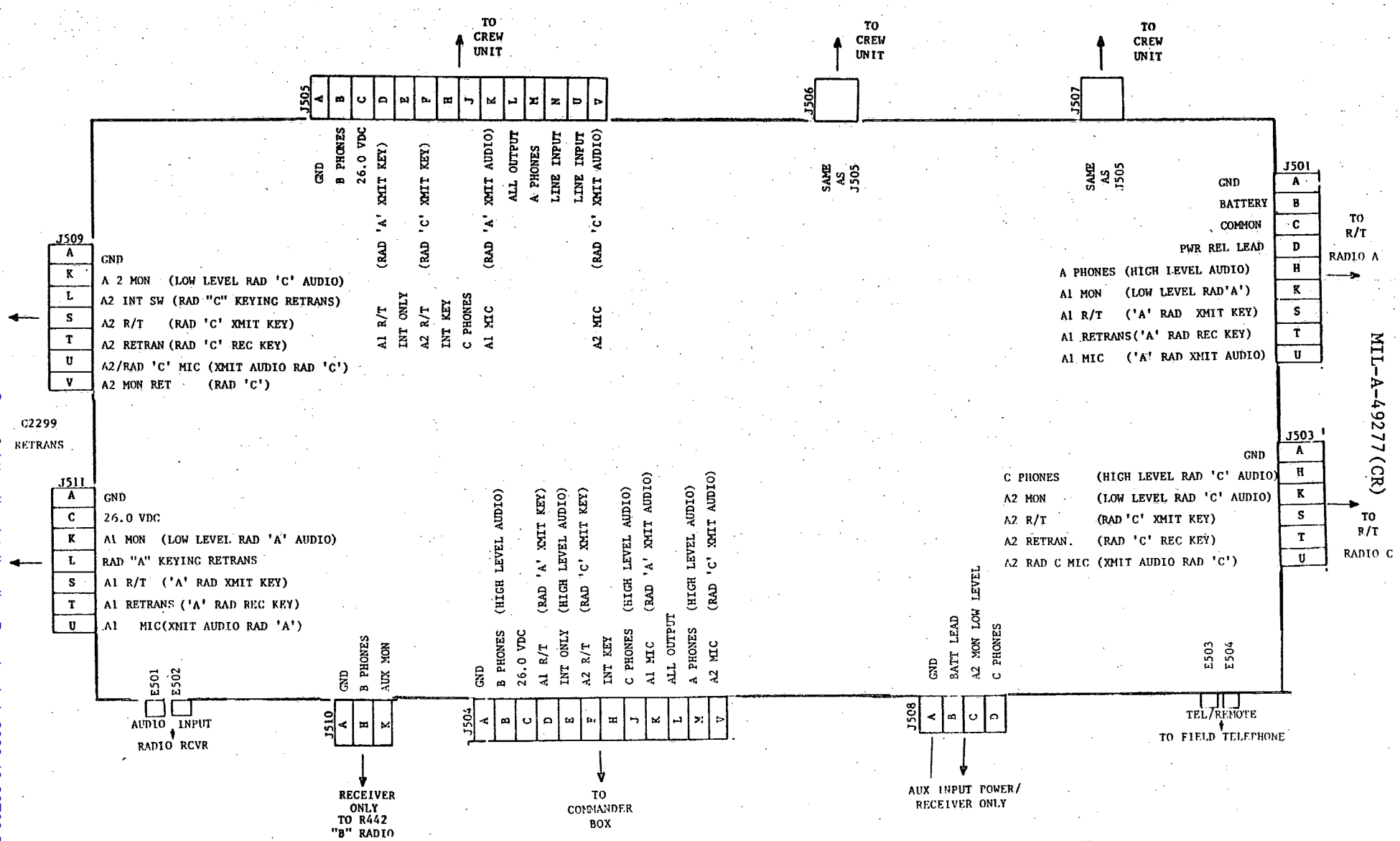


FIGURE 2 AM/1780A/VRC UNIT ELECTRICAL INTERFACE

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. Controls routing of transmitter keying signals.

. Enables unit to obtain power from a radio receiver or power source connected to J508 when no radio receiver/transmitters are used in lieu of the normal power source connected to J501.

3.1.7 Tel/Remote Jacks. A pair of binding posts shall be provided to permit connection of a remotely located telephone or switchboard for communication on intercom circuit.

3.1.8 Audio Input. A pair of binding posts shall be provided to connect audio input from an auxiliary radio receiver (7.35 V rms ±5%).

3.1.9 Interface Definition. The AM-1780A/VRC Amplifier, Audio Frequency unit shall interface with a variety of audio accessories, radio sets (receive-transmit), receivers (radio receive only), and field wire line. When radio sets are used with the AM-1780A/VRC Amplifier, Audio Frequency unit, they are connected through J501/503 (receive-transmit) and J508, J510 (Receive Only). The re-transmission function requires that two radio sets be connected to receptacles J501 and J503. This function serves as a radio relay to increase the range between the two external radio sets.

When radio sets (receive-transmit) are not used, DC power for the AM-1780A/VRC Amplifier, Audio Frequency unit is supplied through receptacle J508.

3.1.10 Main Power Switch. When the Main Power switch is in the Normal position and a receiver-transmitter is connected to J501, power is received through Pins B and C, J501. An auxiliary power input, J508, shall be provided to permit use without radios.

3.1.11 Audio Amplifier Printed Circuit Board Assembly. This assembly contains the signal processing and conditioning portion of the AM-1780A/VRC. From this board signals are routed to and from crew units and radios using solid state relays. This board also contains the line function interface and audio power amplifier which drives the headsets of all the crew units (see 4.12).

3.1.12 DC to DC Converter. This assembly contains a power line conditioner and DC to DC converter which generates all the positive and negative voltages to run the AM-1780A/VRC and all the crew units. This module also provide power line surge and transient suppression and short circuit protection in accordance with 3.8.11 (see 4.13).

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3.1.13 Interconnect Circuitry. The Interconnect Circuitry is composed of two areas:

- Flex circuitry is utilized for the majority of board interconnections in accordance with MIL-P-50884.
- Discrete wiring completes the remaining electrical connections (see 4.13).

3.2 First Article. When specified in the contract or purchase order, the contractor shall furnish thirty-four first article units in accordance with 4.3 (see 6.3.2.4 and 6.5).

3.3 Construction. The Amplifier, AM-1780A/VRC shall be constructed in accordance with the drawings on Data List SM-D-991090, and the requirements specified herein (see 4.4).

3.4 Parts, Materials and Processes, General. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply (see 4.4).

3.4.1 Printed Wiring Boards. Printed Wiring Boards are fabricated in accordance with MIL-STD-275 and MIL-P-55110 (see 4.5.1.1).

3.5 Finish, Protective. The AM-1780A/VRC shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc. and necessary touch-up after mounting (see 4.4). The final paint film on Type I surface shall be final film V in accordance with MIL-F-14072.

3.6 Marking. (See 4.4)

3.6.1 Conformance. The Markings shall conform to MIL-M-13231 with approvals in accordance with requirements specified in the contract or purchase order (see 4.4). Front panel marking shall be Group I as described in the specification.

3.6.2 Visibility of Markings. Whenever practical, parts and assemblies shall be so mounted so that their identification markings will be readily visible with minimum of disassembly of the equipment.

3.6.3 Serial Number. Each AM-1780A/VRC shall be consecutively serial numbered for each contract beginning with the serial numbers provided by the contracting officer (see 4.4).

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3.7 Die Castings. (See 4.4)

3.7.1 Quality of Die Castings. Quality of die castings shall be as follows:

(a) Surface conditions shall be clean, dry and free from chips.

(b) Cast surface finish shall be of a painting grade which allows some streaks and chill marks which can be covered by paint.

(c) Flash must be removed by grinding or filing.

(d) The casting, after impregnation, shall be air tight, and no leakage shall be allowed under a pressure of 10 psi gage.

(e) Flatness shall be as specified in the government furnished drawings. (SM-D-415294, SM-D-415291)

(f) Dimensions shall be as specified in the government furnished drawings. (SM-D-415294, SM-D-415291)

3.7.2 Impregnation of Aluminum Castings. All exterior aluminum castings shall be impregnated in accordance with MIL-STD-276, using Type II impregnating material in accordance with MIL-I-6869. Castings which have been impregnated shall be marked "IMP". The impregnation shall be to correct general seepage only and shall not be used to correct poor foundry techniques or significant porosity (see 4.4).

3.8 Service Conditions. The equipment shall meet the following service conditions (see 4.5.3).

3.8.1 Temperature. The AM-1780A/VRC shall operate in ambient air temperatures of -40°F to +150°F. The equipment shall withstand storage temperatures of -80°F to 160°F (see 4.6.1).

3.8.2 Humidity. (Moisture-Resistance). The equipment shall operate in relative humidity conditions per MIL-STD-810 (see 4.6.2).

3.8.3 Altitude (Elevation). The equipment shall operate at altitude of 0 to 15,000 ft. above sea level (see 4.6.3).

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3.8.4 Vibration. The AM-1780A Amplifier, Audio Frequency shall withstand vibration associated with vehicular transport and the requirements of Method 514, Procedure VIII, using curve W, of MIL-STD-810 (see 4.6.4).

3.8.5 Bounce, Vehicular. The equipment shall pass the bounce requirements of Method 514, Procedure IX, Part 2, of MIL-STD-810 (see 4.6.5).

3.8.6 Shock, Ballistic. The equipment shall withstand the shock test for lightweight equipment in accordance with MIL-S-901 (see 4.6.6).

3.8.7 Fungus. Equipment and assemblies shall show no evidence of visible fungus or corrosion when subjected to 28 days of fungus test in accordance with Method 508 of MIL-STD-810 (see 4.6.7).

3.8.8 Salt Fog. The equipment and assemblies shall be exposed, in their operating configuration, to 48 hours of Salt Fog in accordance with Method 509, Procedure 1, MIL-STD-810. The equipment and assemblies shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface (see 4.6.8).

3.8.9 (U) Nuclear Survivability. The AM-1780A/VRC shall comply with the Nuclear Survivability Requirements, as specified in Appendix "B" (Classified Statement of Criteria) of this specification. Mechanical configurations, Electronic Assemblies, Electronic Equipment, Electronic Circuits and Electronic Components shall withstand the specified Nuclear Environment. If nuclear hardening fixes are deemed necessary to ensure the nuclear survivability of the equipment, the contractor shall implement the hardening fixes and shall demonstrate through the use of appropriate validation techniques that the equipment satisfies the specified nuclear criteria herein. The amplifier shall be capable of meeting the operational test requirements in 4.12, 1 hour before and within 1 hour after being subjected to the specified nuclear environment. This recovery in operational effectiveness may include automatic or manual correction of nuclear environment-induced changes in the system but not the replacement of piece parts or components. If nuclear environment testing is required to verify the nuclear survivability of the production AM-1780A/VRC, the AM-1780A/VRC shall be energized and the contractor furnished equipment (CFE) and the government furnished equipment (GFE) shall be configured for the worst case equipment response during exposure. The contractor shall ensure that the GFE is not degraded by nuclear environment-induced signal, coupled from the CFE; and vice versa, that the GFE is not degraded by nuclear environment-induced signals coupled from the GFE (see 4.6.9).

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3.8.10 Electromagnetic Compatibility. The AM-1780A/VRC shall comply with MIL-STD-461, Part 4, Class A, CE01, CE03, CS01, CS02, CS06, RS02, RS03, RE02 and RE02.1 (see 4.6.10). The AM-1780A/VRC will only be required to pass CS01, RE02, and RE02.1 for acceptance testing (see 4.6.10).

3.8.11 Transients. The Amplifier, Audio Frequency shall comply with MIL-STD-1275, 6.2.3, Spikes Imported Into Equipment and 6.2.4, Voltage Surges Imported Into Equipment on Power and Keying Lines. The transients test will be performed on the power and radio key lines only (see 4.6.11).

3.9 Preconditioning (See 4.7) Each unit shall be preconditioned to ensure proper workmanship and to exclude infant mortality. The equipment shall be capable of meeting operational test requirements in 4.11 before and after the following tests:

3.9.1 Bounce Preconditioning. Each unit will be subjected to bounce preconditioning for time interval of 1 minute as described in 4.7.1.

3.9.2 Burn-in (debugging). Each Amplifier, Audio Frequency shall be subjected to a burn-in operation of 48 hours, at 27.0 ±10% volts DC input. The last 24 hours of the 48 hour burn-in shall be failure free. The equipment shall meet requirements specified in 4.7.2, without subsequent processing after subjection to burn-in procedures.

3.10 Performance Characteristics. The AM-1780A/VRC shall have characteristics as described herein. Unless otherwise specified values set forth to establish the requirements for performance apply to performance under standard conditions and all combinations of the environmental conditions specified in paragraph 3.4. Compliance with the design requirements shall not relieve the contractor of the responsibility of satisfying the performance characteristics specified herein. The AM-1780A/VRC shall have the following standard performance characteristics (see 4.8).

3.10.1 Subassembly. The AM-1780A/VRC has the following subassemblies: A1 (Audio Amplifier) and A2 (DC-DC Converter). The subassemblies shall be tested as specified in Table II of this document (see 4.8.1).

3.10.2 System.

3.10.2.1 Intercom Amplifier Output Impedance. The audio amplifier output impedance measured at J504L shall be between 8 and 30 ohms. The impedance at the line terminals shall be 100 ohms + or - 20 percent (see 4.8.2.1).

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3.10.2.2 Intercom Amplifier Audio Output Power. The audio power output shall be not less than the following: (See 4.8.2.2)

Load Impedance	Minimum Power Output
100 Ohms	1 Watt
600 Ohms	200 Milliwatts

3.10.2.3 Tel/Remote Output. With a 600 ohm load connected to the Tel/Remote terminals there shall be a voltage across the Tel/Remote terminal from .350V RMS to .622V RMS (see 4.8.2.3)

3.10.2.4 Intercom Amplifier Input Impedance. The impedance of each circuit shall be as specified in 4.8.2.4, + or -20%.

3.10.2.5 Input Voltage Range. Amplifier, Audio Frequency shall meet the requirements of this specification when operating from a 22 to 30 volt, DC source (see 4.8.2.5).

3.10.2.6 Audio Response. The audio response of the intercom amplifier shall be within the values indicated below (see 4.8.2.6).

Frequency (Hz)...	150	300	1000	3500	5000
Output (dB).....	-30min	-5to +4	>1 watt (REF)	-5 to +4	-14 min.

3.10.2.7 Distortion. Total harmonic distortion shall not exceed the following: (See 4.8.2.7)

Load Impedance	Distortion
100 Ohms	5 Percent
600 Ohms	10 Percent

3.10.2.8 Separate Control. It shall be possible to turn on power to the interphone amplifier independently of the radio set (see 4.8.2.8).

3.10.2.9 Power Switch. The main PWR switch shall supply relay voltages of 22 Vdc to 30 Vdc, nominally 25.5 Vdc, to the main power relay located in the receiver-transmitter (see 4.8.2.9).

3.10.2.10 Monitor Inputs. The A1 MON (RAD "A") and A2 Monitor (RAD C) shall be connected to the (INT Accent) and amplifier circuitry through the INSTALLATION SWITCH S501 (see 4.8.2.10). There shall be continuity between J501, 11-K and J503, 9-K, J508-C when switch is in INT only or OTHER position.

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3.10.2.11 A2 Monitor. This test shall assure proper operation of the INSTALLATION SWITCH (see 4.8.2.11).

3.10.2.12 Transmit Control. The INSTALLATION SWITCH shall provide a circuit for the control of A1 R/T and A2 R/T from the control boxes (see 4.8.2.12).

3.10.2.13 Radio Silence Control. The RADIO TRANS switch shall control radio transmission capability from the crew control and from the commander's control box (see 4.8.2.13).

3.10.2.14 Audio Transfer. The intercom key shall, when energized, transfer the following control box functions to the output of the intercom amplifier; A Phones, B Phones, C Phones, INT ONLY (see 4.8.2.14).

3.10.2.15 Tel/Remote Transformer Transfer. The intercom key shall, when energized, disconnect the transformer from the input of the intercom amplifier and transfer it to the Tel/Remote output (see 4.8.2.15).

3.10.2.16 INT. ACCENT Switch. The intercom key, when energized, shall reduce the level of the A1 and A2 monitors, AUX MON, and AUDIO IN inputs by 20dB + or -3dB when the INT ACCENT switch is in the on position and the intercom is keyed (see 4.8.2.16).

3.10.2.17 Tel/Remote Function. The Tel/Remote function shall extend to all crew receptacles (see 4.8.2.17).

3.10.2.18 Crew Microphone Transfer. The intercom key, when energized from any crew receptacle or commander's receptacle, shall transfer all crew A1 microphones to the input of the intercom amplifier (see 4.8.2.18).

3.10.2.19 COMMANDER's Microphone Transfer. The intercom key, when energized from any of the crew receptacles, shall not affect the commander's microphone circuit. When it is energized from the commander's receptacles, the commander's microphone shall be connected to the input of the intercom amplifier (see 4.8.2.19).

3.10.2.20 Intercom Radio Isolation. Intercom signals shall not interfere with retransmitted audio. The highest intercom level permitted in the radio monitor circuits is a maximum of 2.2 millivolts (mv) (see 4.8.2.20).

3.10.2.21 Power Supply. Test to verify performance of the Power Supply which provides +26Vdc to the crew units (see 4.8.2.21).

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3.10.2.22 Continuity. Electrical continuity of wiring and of other connections to the housing shall be verified by a point-to-point test which is described in 4.8.2.22.

3.11 Reliability. Each lot of AM-1780A/VRC Amplifiers tested shall demonstrate a specified Mean-Time-Between-Failure (MTBF) of 5,000 hours. The required reliability shall be demonstrated by a reliability test using the reliability test plan in Appendix A for the Amplifier, Audio Frequency AM-1780A/VRC (See 4.5.4)

3.12 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.9.1).

3.13 Air Seal. The equipment case shall provide means for enabling performance of the air-seal test. When tested as specified in 4.10, the decrease in vacuum shall not exceed 0.01 pounds per square inch, gauge, per minute (see 4.10).

3.14 Immersion. The equipment shall show no evidence of leakage after it is tested as specified in 4.11 per MIL-STD-810, Method 512.1, Procedure I (see 4.11).

3.15 Operational Inspection. Each Amplifier, Audio Frequency shall be given a final inspection, as specified in 4.12, after all other inspections and tests are completed, to determine that the unit is operable prior to packaging (see 4.12).

3.16 Technical Literature and Running Spare Parts. Technical literature and running spare parts shall be furnished when required by the contract or purchase order and shall be identical to corresponding parts in the equipment furnished on the order (see 6.2).

3.17 Workmanship. Amplifier, Audio Frequency shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs herein: (See 4.4, 4.13)

a. In MIL-P-11268 the following requirements are applicable:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general

Soldering and brazing
Cleaning of units.
General requirements for securing of parts.
Process for welding.
Clean after welding.
Controls.
Self-tapping screws.
Cleaning prior to welding.

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Shielding on wire and cable.

b. In this specification:

3.5 Finish
3.6 Marking
3.7 Casting impregnation and quality.

c. Conformal Coating per MIL-STD-275.

3.18 Systems Safety Engineering. (See 4.16)

3.18.1 Personnel Hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection (see 4.16).

3.18.2 Edge Rounding Exposed. Exposed edges and corners shall be rounded sufficiently to minimize laceration/puncture hazards, the design goal being a minimum radius of 0.04 inch (1 mm) for edges and a minimum of 0.5 inch (12.7 mm) for corners.

3.18.3 Radioactive Materials. Radioactive materials shall not be used (e.g., luminous dials/marking, electron tubes, surge arrestors and lenses).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. Inspection shall be classified as follows:

(a) First Article Inspection (does not include packaging) (see 4.3).

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(b) Inspection covered by subsidiary documents (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Packaging inspection (see 4.14).

4.3 First Article Inspection. This inspection will be performed by the contractor unless otherwise specified in the contract.

4.3.1 First Article Units. The contractor shall furnish 34 first article units of the AM-1780A/VRC.

4.3.2 First Article Inspection. The First Article Inspection shall consist of inspections specified in Table I and shall be performed in the order specified herein.

4.3.3 First Article Data. The First Article Test Plan and Test Reports shall be as required in the contract or purchase order.

4.4 Inspection Covered by Subsidiary Documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment.

Item	Where Required
Parts, Materials and Process used in Electronic Equipment	3.4
Finish	3.5
Marking	3.6
Casting Impregnation and Quality	3.7.2
Serial Numbers	3.6.3

4.5 Quality Conformance Inspection of Equipment. Each equipment produced under this specification shall be first subjected to bounce preconditioning (see 3.9.1) and then to the 48 hour burn-in (debugging) (see 3.9.2) prior to selection of samples for inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements (see 6.9). Test equipment for Government verification inspection shall be made available by the contractor.

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TABLE I. First Article Inspection (note d)

INSPECTIONS	RQTS PARA.	TEST PARA.	UNIT NUMBER 1	2	3	4	5-34	35-39	NOTES
1 INSP. COVERED BY SUBSIDIARY DOCUMEN	----	4.4							a
2 BOUNCE PRECONDITIO	3.9.1	4.7.1							a
3 ELEC. PRECONDITION	3.9.2	4.7.2							a
4 GROUP A INSPECTION									a
5 GROUP B INSPECTION									a
6 GROUP C INSPECTION									b
LOW TEMP	3.8.1	4.6.1.1	1						d
HIGH TEMP	3.8.1	4.6.1.2	2						d
ALTITUDE	3.8.3	4.6.3	3						d
HUMIDITY	3.8.2	4.6.3		1					d
FUNGUS	3.8.7	4.6.7			1				c
SALT FOG	3.8.8	4.6.8				4			c
VIBRATION	3.8.4	4.6.4				1			d
BOUNCE,VEHICULAR	3.8.5	4.6.5				2			d
SHOCK BALLISTIC	3.8.6	4.6.6				3			d
NUCLEAR SURVIVABIL	3.8.9	4.6.9						1	e
EMI	3.8.10	4.6.10	5						d
TRANSIENTS	3.8.11	4.6.11	4						d
7 GROUP D INSPECTION	3.11	4.5.4					1		

NOTES:

a. Inspections 1 thru 5 shall be performed on all units in the order shown.

b. The unit columns specify the order of inspection by number

c. Before proceeding with subsequent inspection the equipment shall be thoroughly washed, cleaned, dried and refurbished.

d. The order on non-damaging test may be varied, except that the vibration, vehicular bounce and ballistic shock shall be done on the same unit.

e. If nuclear testing is required, units 35-39 shall be used.

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4.5.1 Group A Inspection

4.5.1.1 Group A Inspection of Subassemblies. Each subassembly on contract or purchase order when procured as a spare shall be inspected for conformance as specified in Drawing Nos. SM-A-991154 and SM-A-991155 (see 3.4.1).

4.5.1.2 Group A Inspection of Completed Assembly. Each unit on contract or purchase order shall be inspected for conformance to the inspection specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105, using the general inspection levels and AQLs indicated in Table II. Group A Inspection shall be performed in any order.

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TABLE II. Group A Inspection

Inspection	Req. Para.	Insp. Para.	AQL * Major	AQL Minor
Visual and Mechanical:				
(Subassembly and completed assembly)	3.17 3.18	4.13 4.16	1%	4%
Subassembly Electrical:	3.10.1	4.8.1	1%	4%
System Electrical:				
Separate control	3.10.2.8	4.8.2.8	1%	
Power Switch	3.10.2.9	4.8.2.9	for the group	
A2 Monitor	3.10.2.11	4.8.2.11		
Transmit Control	3.10.2.12	4.8.2.12		
Radio Silence Control	3.10.2.13	4.8.2.13		
Audio Output	3.10.2.2	4.8.2.2		
Tel/Remote Output	3.10.2.3	4.8.2.3		
Distortion	3.10.2.7	4.8.2.7		
Monitor Inputs	3.10.2.10	4.8.2.10		
Audio Transfer	3.10.2.14	4.8.2.14		
Tel/Remote Transformer	3.10.2.15	4.8.2.15		
Int Accent Switch	3.10.2.16	4.8.2.16		
Tel/Remote Function	3.10.2.17	4.8.2.17		
Crew Microphones	3.10.2.18	4.8.2.18		
Commander's Microphone	3.10.2.19	4.8.2.19		
Radio Isolation	3.10.2.20	4.8.2.20		
Power Supply	3.10.2.21	4.8.2.21		
Continuity	3.10.2.22	4.8.2.22		
Air Seal	3.13	4.10		

* Applies after Preconditioning (see 4.7).

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4.5.2 Group B Inspection. This inspection (including sampling) shall conform to Table III and to the procedures for small-sample inspection of Standard MIL-STD-105, using the special inspection levels. Group B inspection shall be performed on inspection lots that have passed Group A inspection.

TABLE III. Group B Inspection

Inspection	Req. Para.	Insp. Para.	AQL
Output impedance	3.10.2.1	4.8.2.1	
Input impedance	3.10.2.4	4.8.2.4	4%
Input Voltage Range	3.10.2.5	4.8.2.5	for the
Audio response	3.10.2.6	4.8.2.6	group
Immersion	3.14	4.11	4%
Interchangeability	3.12	4.9.1	6.5%
(Mechanical Dimensions)		4.9.1(a)	
Electrical		4.9.1(b)	

4.5.2.1 Group B Sampling Plans. The Group B Sampling Plans, for the acceptable quality levels (AQL) listed in Table III shall be as follows:

AQL	Inspection Level
2.5%	S-4
4.0%	S-4
6.5%	S-4

4.5.2.2 Order of Inspection Within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.3 Group C Inspection. This inspection shall be as listed in Table IV, and shall be performed on sample units that have been subjected to and met Group A and Group B inspection. Order of test shall conform to Table IV (see 3.8).

4.5.3.1 Sampling for Inspection of Equipment. Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

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Subgroup I. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every month. Each unit shall be subjected to Temperature (high and low), Vehicular bounce, and Transients.

Subgroup II. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every month. Each unit shall be subjected to Humidity testing.

Subgroup III. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every third month and shall be subjected to Vibration and Ballistic Shock tests, in that order.

Subgroup IV. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every fourth month. One unit shall be subjected to Altitude and Salt Fog testing, in that order. One unit shall be subjected to Fungus testing.

Subgroup V. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every sixth month and subjected to EMI testing.

Subgroup VI. If required, tests shall be performed on the five units numbered 35-39 of the First Article Test units.

TABLE IV. Group C Inspection

Subgroup	Inspection	Requirement Paragraph	Inspection Paragraph
I	Low Temperature	3.8.1	4.6.1
	High Temperature	3.8.1	4.6.1
	Bounce, Vehicular	3.8.5	4.6.5
	Transients	3.8.11	4.6.11
II	Humidity	3.8.2	4.6.2
III	Vibration	3.8.4	4.6.4
	Shock, Ballistic	3.8.6	4.6.6
IV	Altitude	3.8.3	4.6.3
	Salt Fog	3.8.8	4.6.8
	Fungus	3.8.7	4.6.7
V	EMI	3.8.10	4.6.10
VI	Nuclear Survivability	3.8.9	4.6.9

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4.5.3.2 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action (1) the process and materials as applicable, and (2) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance representatives. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Group A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.3.3 Failure Reporting and Analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.5.3.4 Reinspection of Conforming Group C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damages are repaired and the sample units are resubjected to and pass Group A inspection only.

4.5.4 Group D Inspection. The Amplifier, Audio Frequency AM-1780A/VRC shall be selected from production lots that have successfully passed burn-in (debugging) and Group A and B tests of this specification (see 4.5.1 and 4.5.2) and shall be subject to the Reliability Test Plan in Appendix A for the AM-1780A/VRC. For the purposes of reliability testing production lots will be determined as outlined in Appendix A.

4.5.4.1 Sampling for Group D Inspection. Samples shall be selected at random from the first production lot for first article tests. For subsequent Group D inspection, units shall be selected at random from required production lots as described in Appendix A.

4.5.4.2 Group D Failures. A Group D failure is defined as any deviation in performance from the requirements specified in Appendix A.

4.5.4.3 Non-compliance. In case of Group D failure occurrence, procedures and requirements specified elsewhere in contract shall be followed.

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4.5.4.4 Reinspection of Conforming Group D Sample Units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are re-subjected to and pass Group A inspection.

4.5.5 Demonstration of Features. Features and functions shall be demonstrated and/or exercised and shown to be present and/or functional in a qualitative manner. Such demonstrations may be combined with qualitative tests or inspections which such tests or inspections clearly demonstrate or exercise the feature and/or function.

4.6 Service Conditions.

4.6.1 Temperature.

4.6.1.1 Low Temperature. The equipment shall be tested in accordance with Method 502, Procedure 1, of MIL-STD-810. The storage temperature shall be -80°F, storage duration is 24 hours and the operating temperature shall be -40°F. Before and after testing the equipment shall meet full specification performance for the measurements specified in 3.10.2.2 and 3.10.2.7 herein (see 3.8.1).

4.6.1.2 High Temperature. The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be +150°F. Before and after testing the equipment shall be tested and shall meet full specification performance for the measurements specified in 3.10.2.2 and 3.10.2.7 herein (see 3.8.1).

4.6.2 Humidity (Moisture-Resistance) Test. The equipment shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Before and after testing the equipment shall meet full specification performance for the measurements specified in 3.10.2.2 and 3.10.2.7 herein. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling or fails subsequently, it does not pass the test. No repair or replacement of parts shall be permitted (see 3.8.2). The AM-1780A/VRC shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials.

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4.6.3 Altitude (Elevation). The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810. Before and after testing the equipment shall meet full specification performance for the measurements specified in 3.15, Operational Inspection (see 3.8.3).

4.6.4 Vibration. The equipment shall be subjected to vibration testing in accordance with MIL-STD-810, Method 514, Procedure VIII, using curve W. There shall be no physical damage and the equipment shall meet the requirements of Intercom Amplifier Audio Output Power 3.10.2.2 and Distortion 3.10.2.7 herein (see 3.8.4).

4.6.5 Bounce, Vehicular. The equipment shall be tested in accordance with Method 514, Procedure IX, Part 2 of MIL-STD-810. The equipment shall be energized at the end of the 3 hour period, and shall be tested for compliance with 3.10.2.2 and 3.10.2.7 and shall be mounted to the Vehicular Adapter Plate 48 by 48 inch, 305 pound or equal. Accessories shall be selected from those listed on SC-A-46439. There shall be no physical damage. After testing, the equipment shall meet the requirements of Intercom Amplifier Audio Output Power 3.10.2.2 and Distortion 3.10.2.7 (see 3.8.5).

4.6.6 Shock, Ballistic.

(a) Test Method. The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in MIL-S-901. Amplifier, Audio Frequency AM-1780A/VRC mounted on a bracket assembly conforming to Drawing SC-D-48664, shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1 foot blow, 3 foot blow, and 5 foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible (see 3.8.6).

(b) Performance. Upon completion of the above test and before repairs are made, tests shall be performed for compliance with 3.10.2.2 and 3.10.2.7.

4.6.7 Fungus. The equipment and assemblies shall be subjected to 28 days of the fungus test specified in Method 508 of MIL-STD-810. For gasket or other non-hermetically sealed equipment, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The equipment shall be subjected to the test in this open condition (see 3.8.7).

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4.6.8 Salt Fog. The equipment and assemblies shall be exposed in their operating configuration to salt fog in accordance with Method 509, Procedure I, MIL-STD-810 for 48 hours (see 3.8.8).

4.6.9 (U) Nuclear Survivability Test. The AM-1780A/VRC shall be tested for Nuclear Survivability requirements as specified in the U.S. Army Nuclear and Chemical Agency - Nuclear Survivability Criteria for the Product Improved AM-1780A/VRC, 30 June 1982 (see Appendix B) and contractor prepared and government approval test plan (see 3.8.9).

4.6.9.1 (U) Initial Nuclear Radiation Testing. For all Hardness Critical Items (HCI's) of the mature system design, response data shall be taken on representative samples from all qualified vendors and from different lots. Five samples from at least two diffusion or inspection lots are required. Semiconductor piece-part tests shall be performed to part failure or to 10 times the specification levels (whichever occurs first). When the system is tested and failure is shown to be between 2 to 5 times the initial nuclear radiation requirement, the contractor's hardness assurance program shall address the cause of the failure. The Government will not accept a system whose tested response is less than 2 times the specified initial nuclear radiation environments. No nuclear testing will be performed unless design changes occur which effect HCI components.

4.6.9.2 (U) Electromagnetic Pulse. The AM-1780A/VRC shall be subjected to the EMP environment specified in Appendix B. The AM-1780A/VRC shall meet the operational requirements of Section 3 upon completion of the threat level test.

4.6.10 Electromagnetic Compatibility. Test in accordance with MIL-STD-462 to the limits specified in MIL-STD-461, Part 4, Class A, Paragraph 6, CS01, RE02 and RE02.1 (see 3.8.10).

4.6.11 Transients. Perform test per 3.8.11. The AM-1780A/VRC shall be tested after being subjected to the requirements of 3.8.11 for compliance with 3.10.2.2, 3.10.2.6 and 3.10.2.7 before any further testing or evaluation is performed. No damage or degradation shall have occurred.

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4.7 Preconditioning.

4.7.1 Bounce Preconditioning. The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC as made the L.A.B. Corporation, Skaneatales, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning the unit shall not be repaired, cleaned, or otherwise changed prior to subjection to inspection (see 3.9.1).

4.7.2 Burn-in (debugging). The AM-1780A/VRC shall be continuously operated for a period of 48 hours under the following conditions: (See 3.9.2)

- Input voltage 25.5. Vdc ±10%.
- Load at Audio Out 100 ohm ±5%.
- Temperature: 40 degrees C±5 (95 to 113 degrees F)
- Audio Input signal 220 mVrms 1000 Hz signal at the A1 Mic Input.

Intercom key lines shall be energized one minute out of every ten minutes.

While the unit is in this operating mode, vibration shall be performed as follows: 2.2g + or -10% at any non-resonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. This vibration shall be applied for 10 minutes during each hour of equipment operating time. After preconditioning, there shall be no repair, cleaning or other changes prior to acceptance inspection.

Operational tests (see 4.12) will be performed prior to the start of preconditioning and after completion of 48 hours of Burn-In. During Burn-In Preconditioning, apply the standard audio input to A1 MIC at any of the crew receptacles (J505, 6, 7-K) or the Commander's receptacle (J504-K) and energize the Commander's INT key circuit (GROUND J504-H). Measure the power output for compliance with 3.10.2.2. Section 4.8.2.2 does not apply for this test.

If a failure is detected in a module, the module may be replaced or fixed and the Burn-In cycle restarted. Upon completion of Burn-In, each unit will be stamped. The power module and circuit boards shall also be stamped and identified.

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4.8 Performance Characteristics. All tests shall be conducted under the following standard conditions, except as specified for the individual tests:

Temperature.....Room ambient
Humidity.....Room ambient
Input Voltage.....25.5 volts, dc ±10%
Input Voltage, audio.....0.22V, rms ±1%
Audio load impedance.....100 ohms
Audio frequency.....1,000 Hz

4.8.1 Subassembly. The subassembly modules (SM-D-991121 and SM-F-991093) shall be tested in accordance with SM-A-991155 and SM-A-991154.

4.8.2 System.

4.8.2.1 Intercom Amplifier Output Impedance. Under standard test conditions (4.8), measure the output impedance for compliance to 3.10.2.1.

4.8.2.2 Intercom Amplifier Audio Output Power. Operate under standard conditions (4.8) with audio input applied to A1 MON (J501-K). Set INSTALLATION SWITCH to "OTHER" position for proper termination. Measure power output for compliance with 3.10.2.2. Repeat above with the audio input applied to AUX MON pin of (J510-K) and to A2 MON pin of (J503-K). Apply 7.35 volts, rms, at 1,000 Hz, to AUDIO INPUT (E-502) and measure power output for compliance with 3.10.2.2. Apply the standard audio input to A1 MIC at any of the crew receptacles (J505, 6, 7-K) or the commander's receptacle (J504-K) and energize the commander's INT key circuit (GROUND J504-H). Measure the power output for compliance with 3.10.2.2. Apply the standard audio input to A2 MIC at the commander's receptacle (J504-V) and energize the commander's INT key circuit (GROUND J504-H). Measure power output for compliance with 3.10.2.2 at (J504-M, J504-B, J504-J, and J504-E).

4.8.2.3 Tel/Remote Output. Set the INSTALLATION SWITCH to "OTHER" position and operate the unit under standard operating conditions (4.8). Apply the standard audio input to any of the monitor inputs (J501-K, J503-K, J510-K) to produce one watt power output from intercom amplifier (J505-L). Measure Tel/Remote output (E503-E504) for compliance with 3.10.2.3.

4.8.2.4 Intercom Amplifier Input Impedance. Under standard test conditions (4.8), with the INSTALLATION SWITCH to "OTHER" position, the input impedance measured is specified below for compliance with 3.10.2.4. (Refer to Figure 2, AM-1780A/VRC Electrical Interface).

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Input	Impedance
A1 Mon (low level Rad A Audio)	150 Ohms (J501-K)
AUX Mon (low level receive audio)	150 Ohms (J510-K)
A2 Mon (low level radio C audio)	150 Ohms (J503-K)
MIC AUDIO (Rad "A", Rad "C", XMIT Audio)	150 Ohms (J504-K, J505-K, J504-V)
TEL/REMOTE	5000 Ohms minimum (E503-E504)

4.8.2.5 Input Voltage Range. Perform the following tests at 22 and at 30 volts, dc, to determine the conformance to 3.10.2.5:

(a) Intercom Amplifier Audio Output Power (4.8.2.2).
(b) Distortion (4.8.2.7)

4.8.2.6 Audio Response. Amplifier, Audio Frequency shall be operated under standard test conditions (4.8). The output voltage at the specified frequencies shall be measured for conformance to 3.10.2.6. An output of 1,000 milliwatts, at 1,000 Hz, shall be used as the reference level (0 dB) at J504-M, J504-B, J504-J, and J504-E.

4.8.2.7 Distortion. Amplifier, Audio Frequency shall be operated under standard test conditions (4.8). Distortion shall be measured, with 100- and with 600-ohm loads for conformance to 3.10.2.7 at J504-M, J504-B, J504-J, and J504-E.

4.8.2.8 Separate Control. Supply 25.5 volts dc to the battery lead J508-B and turn MAIN PWR switch to INT ONLY. Set INSTALLATION SWITCH to INT ONLY and POWER CKT BKR to ON. The pilot light shall come on to indicate compliance with 3.10.2.8. Set INSTALLATION SWITCH to OTHER; the pilot light shall go off. Set INSTALLATION SWITCH to RETRANS: the pilot light shall stay off.

4.8.2.9 Power Switch. Supply 25.5 volts dc to the battery lead (J501-B) and to the common lead (J501-C) and turn MAIN PWR switch to NORM. Set INSTALLATION SWITCH to INT ONLY and POWER CKT BKR to ON. The pilot light shall be on and 25.5 volts dc shall be available at the power relay lead (J501-D) for compliance with 3.10.2.9.

4.8.2.10 Monitor Inputs. Set INSTALLATION SWITCH to OTHER position and measure the input resistance at 1,000 Hz, of each of the two monitor inputs (A1 MON, A2 MON) for compliance with 3.10.2.10 (refer to 4.8.2.4). Measure for continuity between the circuit breaker and Pin B of J508 in INT ONLY position, with the MAIN PWR switch in NORM position and for an open circuit with the INSTALLATION SWITCH in the other two switch positions.

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4.8.2.11 A2 Monitor Inputs. Set INSTALLATION SWITCH to OTHER, INT ACCENT to OFF and RADIO TRANS to CDR ONLY. Measure continuity between A2 MON, (J503-K) and A2 MON Return (J509-V) for compliance with 3.10.2.11. Set INSTALLATION SWITCH to RETRANS position and measure an open circuit between the same two points.

4.8.2.12 Transmit Control. Set INSTALLATION SWITCH to OTHER position and RADIO TRANS switch in CDR + CREW position. Measure continuity between A1 R/T J501-S and CMDR A1 R/T J504-D, and measure continuity between A2 R/T J503-S and CMDR A2 R/T J504-F, for compliance with 3.10.2.12. Place INSTALLATION SWITCH in RETRANS position and determine that both above circuits are open.

4.8.2.13 Radio Silence Control. Set RADIO TRANS switch to LISTENING SILENCE position and INSTALLATION SWITCH to OTHER position. There shall be no continuity from the crew or commander's receptacles to the radio set receptacles of the A1 R/T or A2 R/T functions. Place RADIO TRANS switch in CDR ONLY position. There shall be continuity of A1 R/T and A2 R/T functions for the commander's receptacle only. Place RADIO TRANS switch in CDR + CREW position. There shall be continuity of A1 R/T and A2 R/T functions for both crew receptacles and commander's receptacle. Without changing switch positions, energize the intercom key by grounding J505-H of the crew receptacles or (J504-H) at the commander's receptacle. The A1 R/T function for the crew shall be interrupted, but for the commander, the A1 R/T function shall remain intact. The A2 R/T function for both the crew and the commander shall remain intact (see 3.10.2.13).

4.8.2.14 Audio Transfer. Set INT ACCENT switch to OFF position and apply sufficient audio into the AUDIO INPUT to obtain 200 mw into a 600 ohm load connect to the ALL output (J505-L). Connect high impedance phones or oscilloscope across load so the frequency can be identified. Apply another audio source of a different frequency and with sufficient power to supply 200 mw into a 600 ohm load to the A phones pin, (J501-H) receptacle. Move the 600 ohm load and phones to the A phones pin of the crew receptacle (J505-M). The A phones source shall be heard. Energize the intercom key by grounding the INT key (J505-H). The audio input source shall be heard and the source at J501-H shall not be heard. Move the 600 ohm load and phone to the B phones pin of the crew receptacles (J505-B) and move the phones source to the B phones input of the (J510-H) receptacle. With the intercom key deenergized, the phones source shall be heard. Energize the intercom key as before. The phones source shall not be heard and the audio input source heard. Move the phone source to the C phones input of (J503-H) receptacle. Move the 600 ohm load and phones to the C phones pin of one of the crew

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receptacles (J505-J). The C phones source shall be heard when the intercom key is both energized and deenergized. Move the 600 ohm load and phones to the C phones pin of commander's receptacle (J504-J). With the intercom key deenergized, the C phones source shall be heard. With the intercom key energized, the audio input source shall be heard. Connect the 600 ohm load and headphones to the ALL pin of the crew receptacle (J505-L). The audio input source only shall be heard with the INT COM key energized or deenergized and A phones, B phones, and C phones inputs connected. Connect the 600 ohm load and headphones to the Int Only pin of crew receptacle (J505-E) and apply a headphone level to A phones, B phones, and C phones inputs. Nothing shall be heard. Energize the intercom key and the audio input source shall be heard (see 3.10.2.14).

4.8.2.15 Tel/Remote Transformer Transfer. With intercom key deenergized, apply 0.44 volts, rms, at 1,000 Hz to Tel/Remote terminals (E503-E504). Connect a 100 ohm wattmeter load to the intercom amplifier output of the ALL pin of the crew receptacles (J505-L) where there shall be minimum output of 1 watt. Energize the intercom key (J505-H). The output shall go to zero. Connect a 600 ohm load to the Tel/Remote terminals (E503-E504). Energize the intercom key. Apply and adjust a 1000 hertz signal to the A1 mic input at any of the crew receptacles (J505-K) such that there is an output of 1.0 watt at the "ALL" pin of any of the crew receptacle. There shall be 0.44 volts rms, +3 dB, -2 dB across the Tel/Remote terminals (see 3.10.2.15).

4.8.2.16 INT ACCENT Switch. Connect a 600 ohm wattmeter load to the ALL pin of crew receptacles (J505-L) and apply 0.22 volts, rms, at 1,000 Hz to the A1 mon input (J501-K). Set INSTALLATION SWITCH to OTHER position with the INT ACCENT switch in OFF position. There shall be no appreciable difference in output power when the intercom key is energized. Place the INT ACCENT SWITCH in the ON position. Energize the intercom key and measure the decrease in power output for compliance with 3.10.2.16. Repeat the above test for the AUX MON input (J510-K), A2 MON input (J503-K) and the audio input (E502) which will require 7.35 volts, rms, at 1,000 Hz.

4.8.2.17 Tel/Remote Function. Repeat the test in 4.8.2.15, except that the point of measurement is at the crew receptacles (J505-N and J505-U) (see 3.10.2.17).

4.8.2.18 Crew Microphone Transfer. Connect a 150 ohm load to A1 mic (J501-U) and 100 ohm wattmeter load to ALL pin of any of the crew receptacles (J505-L, J506-L, J507-L). With the intercom key deenergized, apply 0.22 volts, rms, at 1,000 Hz, to the A1 mic pin of the crew receptacles (J505-K). Measure 0.22 volts, rms, at the 150 ohm load and no

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output at the wattmeter. Energize the intercom key at any of the crew receptacles (J505-H) or the commander's receptacle (J504-H). There shall be no output at the 150-ohm load and there shall be 1 watt at the wattmeter. Repeat above test for the A2 microphone (J505-V) circuit of the Crew. There shall be 0.22 volts, rms, at 150-ohm (J503-U) load and no output at the wattmeter with the crew (J505-H) intercom key either energized or deenergized (see 3.10.2.18).

4.8.2.19 Commander's Microphone Transfer. Connect a 150-ohm load to the A1 mic pin of J501-U, and a 100-ohm wattmeter load to the A11 pin of the crew receptacles (J505-L, J506-L, J507-L). With the intercom key deenergized, apply 0.22 volts, RMS, at 1,000 Hz to the A1 mic pin of the commander's receptacle (J504-K). Measure 0.22 volts, rms, at the 150-ohm load and no output at the wattmeter. Energize the intercom key at the crew receptacles (J505-H). Measure 0.22 volts, rms, at the 150-ohm load and no output at the wattmeter. Energize the intercom key at the commander's receptacle (J504-H). There shall be no voltage across the 150-ohm load and there shall be 1 watt at the wattmeter. Repeat the above test for the commander's A2 microphone circuit (see 3.10.2.19).

4.8.2.20 Intercom-Radio Isolation. Set the INSTALLATION SWITCH to OTHER position and the INT ACCENT switch to the OFF position. Apply 0.22 volts, RMS, at 1,000 Hz, to A1 or A2 mic (J504-K or J504-V) of the CMDR receptacles and energize the intercom key. Measure the ac voltage at the A1 MON (J511-K) and at the A2 MON (J509-V) to determine compliance with 3.10.2.20.

4.8.2.21 Power Supply. Using the test setup shown in Figure 3, the power supply shall meet or exceed the ripple rejection requirements for specified input AC voltages. DC output shall be +26Vdc ±10% for loads from 60 Ohms to 500 Ohms and DC line voltages (V_{in}) from +22 to +30Vdc. (See Figure 3.) (See 3.10.2.21)

4.8.2.22 Continuity. (See 3.10.2.21). Without any other connections, measure for continuity as follows:

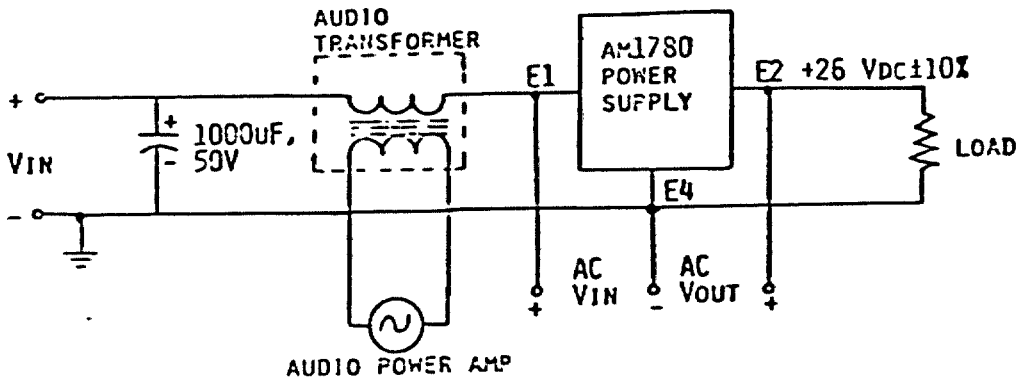
Between	And
Pin A, J501	Pin A of J503, J504, J505, J506, J507, J508, J509, J510, J511 and E501
Pin K, J501	Pin K, J511
Pin S, J501	Pin S, J511

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Between	And
Pin T, J501	Pin T, J511
Pin U, J501	Pin U, J511
Pin H, J503	Pin J, J505; Pin J, J506; Pin J, J507 and Pin D, J508
Pin K, J503	Pin C, J508 and Pin K, J509
Pin S, J503	Pin S, J509
Pin T, J503	Pin T, J509
Pin U, J503	Pin V, J505; Pin V, J506; Pin V, J507 and Pin U, J509
Pin B, J504	Pin B, J505; Pin B, J506; and Pin B, J507
Pin C, J504	Pin C, J505; Pin C, J506; Pin C, J507 and Pin C, J511
Pin E, J504	Pin E, J505; Pin E, J506; and Pin E J507
Pin L, J504	Pin L, J505; Pin L, J506 and Pin L, J507
Pin M, J504	Pin M, J505; Pin M, J506 and Pin M, J507
Pin D, J505	Pin D, J506 and Pin D, J507
Pin F, J505	Pin F, J506 and Pin F, J507
Pin H, J505	Pin H, J506 and Pin H, J507
Pin K, J505	Pin K, J506 and Pin K, J507
Pin N, J505	Pin N, J506 and Pin N, J507 and E503
Pin U, J505	Pin U, J506; Pin U, J507 and E504

Also measure the following for an open circuit to Pin A of all receptacles with switches in all positions;

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FREQUENCY	AC VIN VOLTS RMS	RIPPLE REJECTION $\left(20 \text{ LOG } \frac{\text{AC VIN}}{\text{AC VOUT}}\right)$
50 HZ	2.6	≥ 40 DB
400 HZ	2.6	≥ 40 DB
1 KHZ	2.6	≥ 40 DB
5 KHZ	2.6	≥ 40 DB

FIGURE 3 POWER SUPPLY REQUIREMENTS

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Receptacle	Pins
J501	E,F,J,L,M,N,P,R,T,V
J503	B,C,D,E,F,J,L,M,N,P,R,V
J504	N,P,R,S,T,U
J509	B,C,D,E,F,H,J,M,N,P,R,T
J510	B,D,C,E,F,J
J511	B,E,F,H,J,M,N,P,R,V

4.9 Inspection for Interchangeability.

4.9.1 Inspection for Interchangeability. The mechanical dimensions required by the mechanical gage drawings listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.12. When a listed dimension requirement is not within specified limits, it shall be considered a major defect. The following dimensions or performance shall be gaged or measured:

(a) Mechanical (dimensions). Measurements shall be made at the gage points covered by the drawings listed on the Gage List SC-GL-323191 (see 3.12).

(b) Electrical Interchangeability. All units, subassemblies and modules shall be interchangeable without the need for adjustment. Subassemblies shall comply with the requirements of SM-A-991154 and SM-A-991155.

4.10 Air Seal. Remove POWER indicator light cover to break seal and to make vacuum connection. Immediately thereafter, the equipment, as field transported, shall be subjected to a vacuum of 1 pound per square inch, gage (1 pound per square inch less than the atmospheric pressure surrounding the equipment), applied to the interior of the case. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed 0.01 pound per square inch. The gauge used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound per square inch can be determined readily (see 3.13).

4.11 Immersion. The equipment, as prepared for field transportation, shall be tested in accordance with Method 512.1, Procedure I, of MIL-STD-810 (see 3.14).

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4.12 Operational Inspection. Amplifier, Audio Frequency shall be given an operational check under the standard test conditions specified in 4.8, by keying and unkeying the intercom and checking controlled circuits to ascertain proper operation of switching circuits and correct wiring. System electrical tests in Table II, Group A inspection shall be performed (see 3.15).

4.13 Visual and Mechanical Inspection. Equipment shall be examined for defects listed in MIL-STD-252 and shall meet the requirements of 3.17 herein.

4.14 Packaging Inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging.
b. Quality Conformance Inspection of Packaging.

4.14.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.14.2 Quality Conformance Inspection of Packaging.

4.14.2.1 Materials Inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.14.2.2 Preservation Inspection. Inspection of preservation and interior markings shall be in accordance with Group A and B, Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.14.2.3 Packing Inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table V, "Packing Inspection Provisions". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality Level of 4.0 percent defective.

4.15 Preliminary Quality Acceptance. After satisfactory completion of Reliability Testing per paragraph 4.5.4 an Operational Inspection per paragraph 4.12 will be performed at the contractor's facility. All units shall be tested. After satisfactory completion of this test the units will be shipped to: see shipping instructions outlined elsewhere in this contract.

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TABLE V. Packing Inspection Provisions

No.	Characteristic	Method of Inspection
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

4.16 Safety, Visual and Mechanical Inspection. An inspection shall be performed to verify compliance with those portions of 3.18 (Systems Safety Engineering) which can be determined visually.

4.17 Special Test Equipment. Unless otherwise specified, special test equipment shall be provided for testing by the contractor subject to approval of the Government.

4.18 Standard Test Equipment. Standard test equipment is listed in Table VI by manufacturer and model number. In all cases, equivalent substitutes may be used. Standard test equipment shall be provided for testing by the contractor.

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TABLE VI. Standard Test Equipment

Test Category	Equipment or Equivalent	Description
Burn-in/Bounce Precondi-tioning	HP6012A	60V,50A Power Supply
	Type 1000SC L.A.B. Corporation	Package Tester
	BRFCC-48-1000 L.A.B. Corporation	Vibration Table
	HP8903A	Audio Analyzer
	HP6002A	Programmable Power Supply
Electrical/ Mechanical Performance	HP8903A	Audio Analyzer
	HP6002A	Programmable Power Supply
Environmental and Other Group C Test	L.A.B. Type 1000C	Package Tester. Meets requirements for MIL-STD-810, Method 514.2-IX, Part 2, 4.6.10.2.1
	Fluke 8021B	DMM
	Thermotron M-30-CHM-3-3	Temperature and Humidity Chamber. Meets require-ments of MIL-STD-810, Methods 501.1-II, 502-1-I and 507.1-II.
	HP8903A	Audio Analyzer

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 10-1-1-00-00-NS-X-ED-0-00-A

5.1.2 Level B. 10-1-1-00-00-NS-X-ED-0-00-B

5.2 Packing and Marking. Packing and marking shall be in accordance with MIL-E-55585.

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6. NOTES

6.1 Intended Use. The AM-1780A/VRC Amplifier, Audio Frequency is intended primarily for use in a vehicular communication system requiring intercom facilities. The unit mounts directly to a military vehicle and is normally used with Radio Set AN/VRC-12 ( ), AN/VRC-43 ( ) through -49, AN/VRC-64 ( ), and AN/GRC-160 ( ) vehicular communications systems. The AM-1780A/VRC Amplifier, Audio Frequency is a major component of various installation harnesses (see 1.1).

6.2 Ordering Data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level A or Level B preservation and packing (see Section 5).

(d) Preproduction pack(s) as follows:

Makeup of pack(s)
Number of each kind of pack to be submitted
Inspection to be performed thereon

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required (see 3.15).

(h) Quantity of running spare parts required (see 3.9).

(i) Size data. Size limitations cited herein are based on the dimensions of the equipment for which these requirements were established. Requests to exceed these limitations must be supported by the data indicating why limitations cannot be met and recommend a revised limitation.

(j) When first article rough handling tests are not required.

(k) When first article packaging inspection reports require acquisition activity approval prior to production unit packaging.

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6.3 Definitions.

6.3.1 Equipment Divisions. The terms used herein for equipment divisions conform to MIL-STD-280.

6.3.2 Definitions of Inspection Terms. The inspection terms used herein conform to MIL-STD-109. The following are abbreviated definitions selected from those given therein for immediate reference.

6.3.2.1 Inspection. Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.2.2 Examination. Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.3.2.3 Testing. Testing consists of determinations of compliance, using technical means.

6.3.2.4 Initial Shipments. Initial shipments of end items will be contingent only upon successful completion of all Group C tests (see 3.8).

6.4 Humidity. Natural climatic extremes may include 100-percent relative humidity, with condensation, up to 4 hours at a time. This requirement (see 3.8.2), specified by MIL-STD-810, is a procedure that gives a practical test approximating the natural conditions.

6.5 First Article Inspection. When a first article inspection is required, the items will be tested and should be samples selected from the first production lot. The first article should consist of 39 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

6.6 Group C Inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.2).

6.7 Location of Operational Inspection Test. It is desirable that the operational inspection (4.11) be performed at a location that will minimize handling (which might cause damage to the equipment) after the inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

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6.8 Nomenclature. The contractor shall apply for nomenclature in accordance with the applicable clause in the contract (see 1.2).

6.9 Verification Inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.10 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.11 Level B Preservation. When Level B Preservation is specified, this level of protection will only be used under known favorable condition during transportation, storage and handling.

Custodian:
Army - CR

Preparing Activity:
Army - CR
Project No. 5820-A765

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TABLE VII. Verification Cross Reference Index

METHOD LEGEND
N/A - Not Applicable
1 - Inspection
2 - Review of Analytical Data
3 - Demonstration
4 - Test

TEST LEGEND
A - Group A, Table II
B - Group B, Table III
C - Group C, Table IV
D - Group D
E - First Article
F - Preconditioning

Section 3 Requirement Reference	Verification Methods(s)						Test Category						Acceptance		Section 4 Verification Requirement
	N/A	1	2	3	4	5	A	B	C	D	E	F	Indv	Smpl	
1	X														6.1
1.1															
2.	X														
2.1	X														
2.2	X														
3.	X														
3.1	X														
3.1.1	X														
3.1.2	X														
3.1.3	X														
3.1.4	X														
3.1.5	X														
3.1.6	X														
3.1.7	X														
3.1.8	X														
3.1.9	X														
3.1.10	X														

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	N/A	1	2	3	4	5	A	B	C	D	E	F	Indv	Smpl	
3.1.11		X													4.12
3.1.12		X													4.13
3.1.13		X													4.13
3.2					X									X	4.3, 6.3.2.4, 6.5
3.3			X										X		4.4
3.4	X														4.4
3.5	X														4.4
3.6		X													4.4
3.6.1		X													4.4
3.6.2		X													4.4
3.6.3		X													4.4
3.7		X													4.4
3.7.1		X													4.4
3.7.2		X													4.4
3.8					X				X						4.5.3
3.8.1					X				X						4.6.1

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	N/A	1	2	3	4	5	A	B	C	D	E	F	Indv	Smpl	
3.8.2					X				X						4.6.2
3.8.3					X				X						4.6.3
3.8.4					X				X						4.6.4
3.8.5					X				X						4.6.5
3.8.6					X				X						4.6.6
3.8.7					X				X						4.6.7
3.8.8					X				X						4.6.8
3.8.9			X		X				X						4.6.9
3.8.10					X				X						4.6.10
3.8.11					X				X						4.6.11
3.9												X	X		4.11
3.9.1					X							X	X		4.7.1
3.9.2					X							X	X		4.7.2
3.10	X														4.8
3.10.1					X		X								4.8.1
3.10.2					X										4.8.2
3.10.2.1					X			X							4.8.2.1
3.10.2.2					X		X						X		4.8.2.2

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Section 3 Requirement Reference	Verification Methods(s)						Test Category						Acceptance		Section 4 Verification Requirement
	N/A	1	2	3	4	5	A	B	C	D	E	F	Indv	Smpl	
3.10.2.3					X		X						X		4.8.2.3
3.10.2.4					X			X							4.8.2.4
3.10.2.5					X			X							4.8.2.5
3.10.2.6					X			X							4.8.2.6
3.10.2.7					X		X						X		4.8.2.7
3.10.2.8					X		X						X		4.8.2.8
3.10.2.9					X		X						X		4.8.2.9
3.10.2.10					X		X						X		4.8.2.10
3.10.2.11					X		X						X		4.8.2.11
3.10.2.12					X		X						X		4.8.2.12
3.10.2.13					X		X						X		4.8.2.13
3.10.2.14					X		X						X		4.8.2.14
3.10.2.15					X		X						X		4.8.2.15
3.10.2.16					X		X						X		4.8.2.16
3.10.2.17					X		X						X		4.8.2.17
3.10.2.18					X		X						X		4.8.2.18
3.10.2.19					X		X						X		4.8.2.19
3.10.2.20					X		X						X		4.8.2.20

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3.10.2.21					X		X						X		4.8.2.21
3.10.2.22					X		X						X		4.8.2.22
3.11					X					X					4.5.4
3.12		X													4.9.1
3.13					X		X						X		4.10
3.14					X			X					X		4.11
3.15					X		X								4.12
3.16	X														6.2
3.17		X													4.4, 4.13
3.18		X	X												4.16

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APPENDIX A

RELIABILITY TEST PLAN FOR AM-1780A/VRC

10.0 SCOPE. This test plan defines the methods and procedures to be used to perform the Reliability Testing of the AM-1780A/VRC.

20.0 APPLICABLE DOCUMENTS. The following publications forms a part of this document to the extent referenced herein: MIL-STD-781 "Reliability Tests, EXPONENTIAL DISTRIBUTION".

30.0 REQUIREMENTS. Sample units shall be selected at random from each reliability lot or lots (depending on the frequency of testing required by this specification) in accordance with 4.5.4, shall be subjected to a reliability acceptance test based on a specified mean-time-between-failures (MTBF) of 5000 hours. The total test time per sample unit shall be in accordance with 4.5.2.4 of MIL-STD-781. 5.5 through 5.13 inclusive of MIL-STD-781 shall apply. A reliability testing program shall be conducted in accordance with the following:

a. A First Article Reliability Qualification Test shall be performed on thirty sample units as stated in 3.2, "FIRST ARTICLE", using Test Plan III of MIL-STD-781. θ_o is defined as upper MTBF (5000 hours).

b. Bi-monthly (one test every 2 months) Reliability Acceptance Testing shall commence with the first month of production deliveries. A sample of thirty AM-1780A/VRC, randomly selected from each reliability lot shall be subjected to Test Plan III of MIL-STD-781 with modified accept/reject criteria (see 30.1.7). If each of three consecutively bi-monthly tests, as tested above, has an observed MTBF (total unit "ON" time/number of relevant failures) greater than $2x\theta_o$ the contractor may institute reduced testing in accordance with "C" below, otherwise testing shall continue on a bi-monthly basis as above.

c. Reduced Reliability Acceptance Testing. When requirements for reduced testing specified in "b" above have been met, semi-annual testing (i.e., one Reliability Acceptance Test every six months) using Test Plan VIII of MIL-STD-781 may be initiated. A sample of thirty units selected at random shall represent three consecutive reliability lots. The sample may be selected from the early part of the first month reliability lot of each six months production. If the observed (total equipment "ON" time/number of relevant failures) for an individual test is greater than or equal to 1.75 θ, the Reduced Reliability Acceptance Test procedures described above may be continued. However, if the observed MTBF is less than 1.75θ, bi-monthly reliability testing in accordance with "b" above shall be initiated.

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Reduced inspection may be re-initiated only when conditions of "b" above are fulfilled. Anytime a reject decision occurs when using Test Plan VIII of MIL-STD-781, corrective action shall be taken and the Reliability Acceptance Test shall be repeated using Test Plan III of MIL-STD-781.

30.1 Reliability Test Conditions.

30.1.1 Voltage Cycling. Voltage cycling shall be accomplished as follows: Once every 200 hours of amplifier "ON" time an electrical stress shall be applied to the amplifier. This stress is defined to be a voltage pulse 2 times nominal input voltage in amplitude and a minimum of 10 microseconds in pulse duration. The point in the temperature cycle at which the stress is introduced shall vary during the Reliability Test to conclude application at Step 2 and Step A.

30.1.2 Vibration. Once every hour of testing the units under test shall be subjected to vibration of 2.2 G ±10% at any non-resonant frequency between 20 and 60 Hz measured at the mounting points on the AM-1780A/VRC. This vibration shall be applied for 15 minutes out of each hour of "ON" time.

30.1.3 Humidity. Once every temperature cycle it should be observed and recorded that condensation or frosting occurs. The introduction of humidity shall be a different time and temperature cycles such that the temperature at which condensation or frosting occurs varies during the test.

30.1.4 Temperature Cycle. The testing shall be conducted on a 24-hour, 7 days per week basis using the following steps (Ambient Temperature hours (Step 3 and 7) may be shifted as required in order to complete performance testing):

STEP 1: With the amplifiers off and the chamber temperature at ambient, lower the chamber temperature to -57°C ±2°C. Maintain this chamber temperature for 1 hour.

STEP 2: Raise the chamber temperature to -32°C ±2°C at a rate of temperature change not less than 5°C/minute. Maintain the chamber temperature at -32°C ±2°C for a period of 10 hours. Turn the amplifier on when temperature has reach -32°C ±2°C, see 40.7, audio power output is measured when required by 40.7.

STEP 3. While the amplifier is still "ON", raise chamber temperature to room ambient temperature and maintain for 8 hours. After temperature has stabilized perform Group A Electrical Test, see 40.7.

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STEP 4. Raise the chamber temperature to 55°C ±2°C at a rate of temperature change not less than 5°C/minute. With the amplifier still "ON" maintain chamber temperature at 55°C ±2°C for a period of 15 hours, see 40.7. Audio power output is measured when required by 40.7.

STEP 5. Turn the amplifier "OFF" and raise the chamber temperature to 71°C ±2°C. Stabilize and maintain this chamber temperature for a period of one hour.

STEP 6. Lower the chamber temperature to 55°C ±2°C and turn the amplifier "ON" when temperature has reached 55°C ±2°C. With the equipment "ON", maintain the chamber temperature at 55°C ±2°C for a period of 10 hours, see 40.7, audio power output is measured when required by 40.7.

STEP 7. With the equipment still "ON", lower the chamber temperature to room ambient and maintain this temperature for 8 hours. After temperature has stabilized, perform Group A Electrical Test, see 40.7.

STEP 8. With the equipment still "ON", lower the chamber temperature to -32°C ±2°C at a rate of temperature change not less than 5°C/minute. Maintain this chamber temperature for a period of 15 hours, see 40.7, audio power output is measured when required by 40.7.

STEP 9. Turn the equipment "OFF" and lower the chamber temperature to -57°C ±2°C and maintain for 1 hour.

Repeat Steps 2 through 9 throughout the Reliability Test. Completion of Steps 2 through 9 shall herein be referred to as one temperature cycle.

30.1.5 Duty Cycle. During the equipments on time the duty cycle shall be continuous.

30.1.6 Failure Reporting and Analysis. A failure reporting and analysis system shall be implemented.

30.1.7 Accept-Reject Criteria. The Accept/Reject Criteria shall either be Test Plan III or Test Plan VIII of MIL-STD-781 depending on the phase of testing being conducted in accordance with 30.0 provisions. When Test Plan III is applicable, the maximum allowable number of failures is six. If a reject decision has not been made within the test constraints given in MIL-STD-781, the units shall be considered to pass the test.

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APPENDIX A

40.0 PERFORMANCE

40.1 Validation Measurement. At least once daily the performance parameters of 40.7.1 shall be measured with the equipment stabilized at the temperatures applicable in 30.1.4. If the value of any parameter is not within tolerance, a failure shall be recorded and the required action shall be taken. The failure shall be presumed to have occurred immediately after the last successful measurement.

40.2 Reliability Test Conditions. The Reliability Test shall be conducted in a Test-Fix-Test manner. Upon the occurrence of a failure an analysis shall be performed to determine the failure cause. Any corrective action shall be fully documented. Any design change, modification or replacement shall be applied in like manner to all amplifiers under test at the earliest possible moment after verification of the proposed change and to all amplifiers before delivery. The absence of one amplifier will not affect the ability to continue the test. The test conditions and procedures for the Reliability Test shall be otherwise stated in this specification.

40.2.1 Test Facilities. Test facilities shall be capable of maintaining the conditions specified for the applicable environments and of measuring the amplifier characteristics to the specified accuracy for the duration of the test.

40.2.2 Test Chambers. Test chambers shall be capable of maintaining the ambient and forced air temperatures as specified with an accuracy of ±2°C during the test. Chamber temperatures shall be monitored continuously.

40.3 Voltage Cycling. The AM-1780A/VRC shall be tested for conformance to 30.1.1. Prior to testing the AM-1780A/VRC shall meet full specification performance for the measurements specified in 30.1.1. During testing disconnections shall be made where necessary to protect circuit elements.

40.4 Vibration. A vibration table, located within the chamber, will provide the vibration of 2.2 G ±10% at any non-resonant frequency between 20 and 50 Hz measured at the mounting points on the equipment. The vibration table will be energized for 15 minutes out of each hour of "ON" time. The amplifier shall be hard mounted, no shock isolators shall be used. Vibration shall be normal to the plane of the majority of printed circuit cards. The vibration equipment shall be checked for proper operation each 24 hours of operation.

40.5 Humidity. The AM-1780A/VRC shall be tested for compliance with 30.3. Do not move the amplifier from the chamber for measurement.

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40.6 Temperature Cycle. A temperature cycle is defined as Steps 2 through 9, see paragraph 30.1.4, and consists of a total time period of 72 hours. The first half of the temperature cycle goes from cold soak to low temperature to ambient to high temperature to hot soak. The second half of the temperature cycle goes from hot soak to high temperature to ambient to low temperature to cold soak.

40.7 Performance Measurement. During each temperature cycle, performance measurements will be made either during Steps 2, 3, and 4 or during Steps 6, 7 and 8. During the first temperature cycle, measurements will be made in Steps 2, 3, and 4. Measurements in Steps 6, 7 and 8 shall not be performed during temperature cycle one. During temperature cycle two, measurements will be made in Steps 6, 7 and 8. Measurements in Steps 2, 3 and 4 shall not be performed during temperature cycle two. The measurements shall be made on this alternate temperature cycle basis throughout the Reliability Testing.

40.7.1 Performance Test Perform the following test:

Separate Control	4.8.2.8
Power Switch	4.8.2.9
A2 Monitor	4.8.2.11
Transmit Control	4.8.2.12
Radio Silence Control	4.8.2.13
Audio Output	4.8.2.2
TEL/REMOTE Output	4.8.2.3
Distortion	4.8.2.7
Monitor Inputs	4.8.2.10
Audio Transfer	4.8.2.14
Tel/Remote Transformer	4.8.2.15
Int Accent Switch	4.8.2.16
Tel/Remote Function	4.8.2.17
Crew Microphones	4.8.2.18
Commander's Microphone	4.8.2.19
Radio Isolation	4.8.2.20

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APPENDIX A

40.8 Failure Reporting and Analysis.

40.8.1 Failure Reporting and Documentation.

40.8.1.1 Each failure occurring during the Reliability Test shall be reported and analyzed; and have corrective action taken.

40.8.1.2 The failure shall be documented on a failure report from which will describe the method used to detect the failure. The failure mode and the repair necessary to make the AM-1780A/VRC operational.

40.8.1.3 Each failure will be documented on the appropriate log of failures which contain (in part) the accumulated operating time, at the time of failure, and the failure number.

40.8.1.4 The accumulated operating time on all AM-1780A/VRC units shall be recorded on the operating time log when a failure occurs.

40.8.1.5 When a AM-1780A/VRC is removed from the heat chamber for repair a failure tag will be attached to the unit showing (in part), the failure report number, associated reports and the operation sheet page and line number.

40.8.1.6 A failure detected during performance measurement tests will have the data entry circled on the operation sheet.

40.8.1.7 A failed AM-1780A/VRC will be sent to troubleshooting and repair, where the AM-1780A/VRC shall be analyzed to isolate and repair the failure. The findings will be documented on the failure report and the repair sheet.

40.8.1.8 When a failed AM-1780A/VRC is returned to test, the elapsed operating time meter readings shall be recorded on the operating time log and the date of return recorded on the log of failures.

40.8.2 Failure Analysis. Each failure occurring during the Reliability Test shall be analyzed and classified in accordance with 5.6.2 of MIL-STD-781.

40.8.3 Corrective Action. A corrective action system shall be implemented in accordance with 5.9 of MIL-STD-781.

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40.8.4 Failure Summary and Analysis. The failure summary and analysis shall comply with 5.13.2.1 of MIL-STD-781.

40.9 Reliability Test Report. The Reliability Test Report shall comply with 5.13 of MIL-STD-781. Bi-weekly reports of performance shall also be submitted.

40.10 Accept-Reject Criteria. The decision to reject, continue testing or to accept the Reliability Test shall be determined by either Test Plan III or Test Plan VII of MIL-STD-781, depending on the phase of testing being conducted in accordance with 30.0.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-49277	Amplifier, Audio Frequency AM-1780A/VRC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*. _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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